Mechanism-Inspired Aggregation for Multi-Agent Alpha Discovery: Optimizing Agent Distributions in Heterogeneous LLM Markets

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Abstract

We study multi-agent stock selection as a mechanism-inspired adaptive aggregation problem: a principal-like optimizer allocates aggregation weight across heterogeneous LLM-powered investor types to maximize predictive alpha. Responding to the need for clearer benchmarks in strategic AI systems, we provide a transparent empirical stress test of the MASS (Multi-Agent Simulation Scaling) framework across optimizer choices, agent scales, markets, seeds, and simple controls. At 64 agents on a China A-share pool, the original simulated-annealing baseline remains strongest among the tested aggregation variants (10d Rank IC 0.033 ± 0.010 vs. 0.023 ± 0.006 and 0.021 ± 0.004 for CMA-ES variants). At 512 agents, CMA-ES improves over matched simulated annealing on average in both China (0.028 ± 0.021 vs. 0.011 ± 0.015) and S&P 500 (0.0139 ± 0.0067 vs. 0.0095 ± 0.0063), but the margins are noisy and do not establish a monotonic scaling law. Component and simple non-LLM controls show that consensus is often the stable positive component, while low-disagreement terms can suppress useful signal. These results position adaptive aggregation over heterogeneous LLM agents as an empirical testbed for mechanism-design-inspired questions, while stopping short of formal equilibrium or strategy-proofness claims.

1. Introduction

Large language models (LLMs) are increasingly deployed as autonomous agents in complex decision-making environments, including financial forecasting and trading [6, 9, 10]. A central question in multi-agent LLM systems is how a coordinating procedure should aggregate heterogeneous agent outputs when agent quality, disagreement, and market regimes vary over time. This question is closely aligned with the NExT-Game workshop themes of adaptive systems, game optimization, and benchmarks for strategic AI, but it is also easy to overstate: not every leader-follower pipeline is a formal mechanism-design model.

We study this question in the context of MASS (Multi-Agent Simulation Scaling) [3], a framework for portfolio construction that simulates a population of heterogeneous LLM-powered investor types. Each type receives different market data modalities and generates investment signals; a backward optimizer then adjusts aggregation weights over agent types to maximize predictive Rank IC. We view this system through a *Stackelberg-style* lens: a principal-like optimizer selects an aggregation rule, agents generate decisions from local views, and the optimizer adapts using realized predictive performance.

This is not a classical mechanism-design setting. The LLM agents do not strategically choose reports in response to incentives, and we do not prove equilibrium, welfare, incentive-compatibility, or strategy-proofness guarantees. Instead, we use mechanism-design terminology in a limited empirical sense: a coordinating principal controls an aggregation rule over heterogeneous agents, and the question is how design choices in that rule affect performance. This distinguishes our study from classical mechanism design [8] and recent work on mechanism design for LLM-generated content [2], where incentive properties are central.

The key aggregation idea in MASS is the market disagreement hypothesis: consensus among agents predicts future returns, while disagreement predicts lower returns [1, 7]. MASS forms a signal $S = \alpha \cdot \text{consensus} - (1-\alpha) \cdot \text{disagreement}$ and uses simulated annealing (SA) [5] to optimize agent-type weights. We ablate four aggregation-design choices: objective alignment, learnable α, optimizer capability (SA vs. CMA-ES [4]), and stability regularization.

Our contribution is a transparent empirical benchmark rather than a formal theory paper. We provide: (1) a mechanism-inspired adaptive aggregation framing with explicit limits; (2) prompt and protocol details for reproducibility; (3) multi-seed 64-agent ablations showing that plausible fixes can degrade performance; (4) matched 512-agent China and S&P 500 comparisons showing that CMA-ES improves over matched SA on average, but with high variance and modest margins; and (5) simple and component-level controls that expose when aggregation choices suppress useful consensus signal.

2. Method

2.1. Adaptive Aggregation Formulation

The MASS system can be described as a repeated adaptive aggregation process with a leader/follower ordering:

- **Leader (Principal)**: At each trading day t, chooses nonnegative aggregation weights $\mathbf{d}_t$ over n agent types, plus aggregation weight $\alpha \in [0, 1]$ and stability parameter $\lambda \geq 0$. The weights are normalized inside the scoring rule.
- **Agents**: Each agent type i generates investment signal $s_{t,i}$ based on its data modality and investment style. In our experiments these agents are heterogeneous signal generators, not strategic followers optimizing against the principal's rule.
- **Aggregation**: The combined signal is $S_t = \alpha \bar{s}_t - (1-\alpha)\sigma_t$, where $\bar{s}_t$ is the consensus (weighted mean) and σ_t is the disagreement (weighted standard deviation).
- **Payoff**: The principal's objective is $\text{RankIC}(S_t, R_{t+\Delta})$, the Spearman correlation between the signal and future returns at horizon Δ.

Unlike classical Stackelberg games, we do not model best responses or equilibrium selection. The Stackelberg-style terminology refers only to the order of operations.

2.2. Aggregation-Rule Design Choices

We evaluate four plausible modifications of the MASS optimizer. **Objective alignment**: optimize Rank IC of the final combined signal rather than only the low-disagreement component. **Learnable** α: optimize the consensus-disagreement tradeoff instead of fixing it at 0.5. **Optimizer capability**: replace SA's random pairwise perturbations with CMA-ES, which adapts a covariance matrix over the search distribution. **Stability constraint**: penalize day-to-day changes in agent-type weights via $\text{fitness} = \text{RankIC} - \lambda \|\mathbf{d}_t - \mathbf{d}_{t-1}\|_1/2$.

2.3. Agent Protocol

All ablations use the same prompt family and differ only in the aggregation optimizer. Each agent first emits a JSON investment style containing fields such as risk appetite, holding period, rationality, and stock-pool selector. It then receives a table-form candidate universe and must output exactly n^{sel} legal stock codes under the JSON key "Stock". The system prompt requires JSON-only output. We include a prompt excerpt in Appendix A.

3. Experiments

3.1. Setup

We report 10-day forward return Rank IC as the primary metric. The main 64-agent ablation uses the China 'ih' pool from 2022-12-02 to 2023-06-15, evaluated from 2023-01-02 after a one-month warmup. Each run uses 8 investor types $\times$ 8 agents per type, $n^{sel} = 5$ selected stocks per agent, and a 20-stock candidate universe. We then run matched 512-agent comparisons with 32 investor types $\times$ 16 agents per type. For S&P 500, we use the 2025-10-01 to 2026-03-31 short window with a 40-stock candidate universe for coverage and the same $n^{sel} = 5$. China matched seeds are $\{00, 11, 22\}$ and S&P 500 matched seeds are $\{42, 72, 92\}$. Seeds change LLM sampling paths and random candidate-pool construction.

3.2. China 64-Agent Ablation

Table 1 shows the China 64-agent ablation. The simple original MASS-style baseline remains strongest on the 3-seed mean; adding objective alignment and learned α can be harmful when the optimizer is insufficient, and CMA-ES variants do not beat the original baseline at this scale.

Table 1: China 64-agent ablation results. The original MASS-style baseline remains strongest on the 3-seed mean.

Config	Fitness	α	Optimizer	Seeds	10d Rank IC
A (baseline)	Low-disagree	0.5 fixed	SA	3	$\mathbf{0.033 \pm 0.010}$
B	Combined	0.5 fixed	SA	1	0.007
C	Combined	Learned	SA	1	-0.019
D	Combined	Learned	CMA-ES	3	0.023 ± 0.006
F	Combined	Learned	CMA-ES, λ=0.01	3	0.013 ± 0.026
E	Combined	Learned	CMA-ES, λ=0.1	3	0.021 ± 0.004

The weak stability penalty is particularly unstable: Config F has one strong seed and two weak seeds. This supports a practical lesson rather than a theorem: expanding the aggregation search space without enough optimizer capacity or appropriate regularization can make performance worse.

3.3. Matched 512-Agent Comparisons

We completed matched 3-seed comparisons for both China and S&P 500. Table 2 shows that CMA-ES improves over matched SA on mean Rank IC in both markets, but the margins are noisy and do not establish a monotonic scaling law. In China, E-512 remains below the 64-agent A baseline mean

(0.028 vs. 0.033), so the evidence supports cautious matched-baseline improvement rather than a claim that larger scale alone solves the aggregation problem.

Table 2: Matched 512-agent full-model comparisons. CMA-ES improves over matched SA on mean Rank IC in both markets, but margins are noisy.

Market	Config	Seeds	10d Rank IC	Read
China 'ih'	E-512 (CMA-ES)	3	0.028 ± 0.021	ahead, high variance
China 'ih'	A-512 (SA)	3	0.011 ± 0.015	weaker mean
S&P 500	E-512 (CMA-ES)	3	0.0139 ± 0.0067	ahead, modest gap
S&P 500	A-512 (SA)	3	0.0095 ± 0.0063	competitive

The seed-level ordering is mixed. In S&P 500, A-512 beats E-512 on one of three seeds. Thus the matched result supports optimizer choice as relevant at larger scale, but not decisive dominance.

3.4. Controls and Diagnostics

To evaluate against simpler alternatives, Table 3 adds no-new-LLM controls. Random ranking is near zero in both markets. Momentum and reversal are weak on S&P 500; in China, 20-day reversal is a strong simple control. Component diagnostics show that consensus is often positive while low-disagreement is negative, which helps explain why an aggregation rule can suppress useful signal.

Table 3: Simple and component controls. Low-P/B is excluded from this table pending point-in-time audit.

Control	China 10d Rank IC	S&P 500 10d Rank IC
Random ranking	0.004 ± 0.010	0.001 ± 0.004
5d momentum	-0.004	0.002
20d momentum	-0.051	0.003
5d reversal	0.004	-0.002
20d reversal	0.051	-0.003
A-512 consensus comp.	0.034 ± 0.006	0.039 ± 0.002
A-512 low-disagree comp.	-0.039 ± 0.006	-0.035 ± 0.003

We also computed value-style China controls, but exclude them from headline claims pending point-in-time audit of the corresponding fields. This illustrates why simple financial controls are essential for evaluating LLM-agent trading systems.

4. Discussion

Adaptive aggregation is fragile. More sophisticated mechanisms do not automatically improve performance. At 64 agents, the original SA baseline is strongest among our tested MASS variants, while the China 20-day reversal control is also competitive. At 512 agents, CMA-ES improves over matched SA on average, but high variance and modest SP500 margins argue against a strong scaling-law claim.

Component diagnostics matter. Across the settings we inspected, consensus is often the stable positive component, while the low-disagreement component is negative. The original MASS aggregation can therefore underuse a useful consensus signal if the optimizer or α choice is poorly matched to the data regime.

Evaluation discipline for LLM agents. Our own conclusions changed after adding seeds: the initial single-seed estimate suggesting that E-512 beat the 64-agent baseline did not survive the 3-seed China mean. We recommend reporting seed counts, variance estimates, simple controls, and component diagnostics as standard practice for LLM-agent market studies.

Limitations. The study remains preliminary. We evaluate Rank IC rather than a transaction-cost-aware portfolio backtest; the windows are short; and the mechanism-design framing is descriptive rather than formal. Some simple financial baselines, especially value-style fields such as P/B, require point-in-time audit before headline use. Additional model-sensitivity and cost-scaled replications are left for future work.

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Appendix A. Prompt and Evaluation Details

The MASS prompt protocol is two-stage. First, an agent receives modality-specific feature descriptions and optional macro text, then emits a JSON investment style with fields such as `Risk Appetite`, `Holding Period`, `Strategy Consistency`, `Rationality`, and `StockPoolSelector`. Second, the agent receives a table of candidate stocks and must output exactly n^{sel} legal stock codes under the key `Stock`. The system prompt is: "Strictly follow the user's input prompt and output the result in JSON format as specified. Do not include any additional content."

All reported Rank IC values are daily cross-sectional Spearman correlations between the signal and the relevant forward-return label, averaged over evaluation dates. Random baselines use ten independent random rankings. Full result files and run suffixes are tracked in `EXPERIMENTAL_RESULTS.md`.